

# PE-EC702B — Digital Image and Video Processing

Semester VII • 3L:0T:0P • 3 Credits

## Module 7 Exam Notes: Fundamentals of Video Coding

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## 1. Inter-Frame Redundancy

A video sequence is a series of consecutive images (frames) captured over time.

### Temporal Redundancy

**Inter-Frame (Temporal) Redundancy:** Refers to the massive statistical correlation between successive frames. Since video frames are captured at high rates (typically 24 to 60 frames/sec), adjacent frames contain nearly identical background and object structures, with only minor spatial displacements.

## 2. Motion Estimation and Compensation

Motion estimation detects displacements of objects between frames, while motion compensation encodes only the difference (prediction residual) and the motion displacement vectors.

### 2.1 Block Matching Algorithm (BMA)

The standard method for motion estimation.

- The current frame is divided into non-overlapping macroblocks of size  $N \times N$  (typically  $16 \times 16$ ).
- For each block, we search for the best matching block in a reference frame (past or future) within a search range  $[-p, p]$ .
- **Search Metrics:** Used to measure block similarity. The most common is the **Mean Absolute Difference (MAD)**:

$$\text{MAD}(i, j) = \frac{1}{N^2} \sum_{x=0}^{N-1} \sum_{y=0}^{N-1} |F_t(x, y) - F_{t-1}(x + i, y + j)| \quad (1)$$

where  $F_t$  is the current block and  $F_{t-1}$  is the reference block shifted by displacement  $(i, j)$  (Motion Vector).

### 2.2 Search Strategies

- **Full Search (Exhaustive Search):** Evaluates MAD at every possible integer pixel shift within  $[-p, p]$ . Guarantees finding the global minimum, but is computationally extremely heavy.
- **Three-Step Search (TSS):** A fast, logarithmic search strategy:
  - a. Start at center, evaluate 9 search locations with a coarse step size  $S = \lfloor p/2 \rfloor$ .
  - b. Move center to the location yielding minimum MAD. Reduce step size to  $S = S/2$ .
  - c. Evaluate 8 new locations around the new center. Repeat until step size is 1.

*Computational savings: Requires only 25 computations for  $p = 7$ , compared to 225 for Full Search.*

### 3. Frame Classification and Sequence Hierarchy

Video coding standards classify frames into three types to optimize prediction efficiency:

- **I-Frames (Intra-Coded):** Coded independently of other frames using still-image compression (like JPEG). Acts as random access points. Low compression.
- **P-Frames (Predicted):** Coded using forward motion prediction from a previous reference frame (I or P). Moderate compression.
- **B-Frames (Bidirectional Predictive):** Coded using both past and future reference frames (forward and backward prediction). Yields the highest compression ratios.

#### Video Sequence Structure

**Group of Pictures (GOP):** The sequence structure of I, P, and B frames. A typical GOP pattern is:

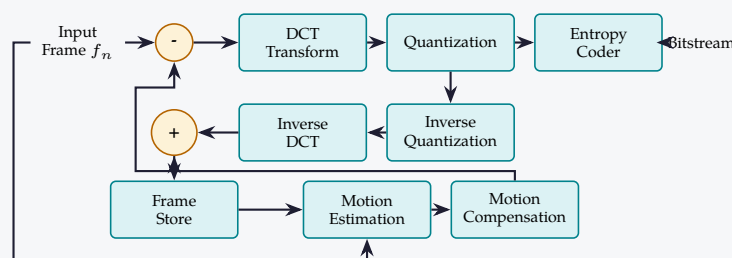
**I B B P B B P B B I**

**Hierarchy:** Video Session  $\rightarrow$  GOP  $\rightarrow$  Picture/Frame  $\rightarrow$  Slices  $\rightarrow$  Macroblocks ( $16 \times 16$ )  $\rightarrow$  Blocks ( $8 \times 8$ ).

### 4. Hybrid Video Encoder and Standards

Modern video compression standards use a hybrid framework combining DPCM predictive loop (motion estimation) with spatial transform coding (2D DCT).

#### Hybrid Video Encoder Block Diagram



#### 4.1 Video Standards

- **MPEG-1 / MPEG-2:** Formats used for VCDs (*MPEG* – 1) and DVDs/Digital TV (*MPEG* – 2).
- **H.264/AVC:** Incorporates intra-frame spatial prediction, quarter-pixel motion vector accuracy, and variable block sizes ( $16 \times 16$  down to  $4 \times 4$ ).
- **H.265/HEVC:** Replaces macroblocks with Coding Tree Units (CTUs) of size up to  $64 \times 64$ , reducing bitrates by 50% compared to H.264 for equivalent quality.

## 5. Worked Examples

### Motion Vector Estimation

**Problem:** Given a  $2 \times 2$  macroblock in the current frame  $F_t$  and its corresponding search region in the reference frame  $F_{t-1}$ . Calculate the Mean Absolute Difference (MAD) for candidate displacement vectors  $\mathbf{v}_1 = (0, 0)$  and  $\mathbf{v}_2 = (0, 1)$ , and identify the optimal motion vector.

**Current Block  $F_t$ :**

$$F_t = \begin{bmatrix} 80 & 85 \\ 70 & 75 \end{bmatrix}$$

**Reference Search Region  $F_{t-1}$ :**

|          | $x = -1$ | $x = 0$   | $x = 1$   |
|----------|----------|-----------|-----------|
| $y = -1$ | 78       | 82        | 88        |
| $y = 0$  | 72       | <b>80</b> | <b>84</b> |
| $y = 1$  | 65       | <b>71</b> | <b>76</b> |

**Solution:**

(a) For  $\mathbf{v}_1 = (0, 0)$ : The candidate block in  $F_{t-1}$  starting at  $(0, 0)$  is:

$$\begin{bmatrix} 80 & 84 \\ 71 & 76 \end{bmatrix}$$

Compute the absolute differences:

$$\text{Diff} = \begin{bmatrix} |80 - 80| & |85 - 84| \\ |70 - 71| & |75 - 76| \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$$

$$\text{MAD}(0, 0) = \frac{0 + 1 + 1 + 1}{4} = \frac{3}{4} = 0.75$$

(b) For  $\mathbf{v}_2 = (0, 1)$  (**vertical shift by  $j = 1$** ): The candidate block in  $F_{t-1}$  shifted by  $(0, 1)$  is:

$$\begin{bmatrix} F_{t-1}(0, 1) & F_{t-1}(1, 1) \\ F_{t-1}(0, 2) & F_{t-1}(1, 2) \end{bmatrix} = \begin{bmatrix} 71 & 76 \\ \text{N/A} & \text{N/A} \end{bmatrix}$$

Since this candidate block contains out-of-bounds elements (N/A), we cannot calculate a valid MAD for  $\mathbf{v}_2 = (0, 1)$ .

**Alternative Candidate Analysis:** Let us analyze displacement  $(i, j) = (-1, -1)$  (shifted top-left by 1 pixel) as an alternative candidate:

- The candidate block in  $F_{t-1}$  for  $(i, j) = (-1, -1)$  is:

$$\begin{bmatrix} F_{t-1}(-1, -1) & F_{t-1}(0, -1) \\ F_{t-1}(-1, 0) & F_{t-1}(0, 0) \end{bmatrix} = \begin{bmatrix} 78 & 82 \\ 72 & 80 \end{bmatrix}$$

- Compute the absolute differences with the current block  $F_t$ :

$$\text{Diff} = \begin{bmatrix} |80 - 78| & |85 - 82| \\ |70 - 72| & |75 - 80| \end{bmatrix} = \begin{bmatrix} 2 & 3 \\ 2 & 5 \end{bmatrix}$$

- Calculate the MAD:

$$\text{MAD}(-1, -1) = \frac{2 + 3 + 2 + 5}{4} = \frac{12}{4} = 3.00$$

**Conclusion and Motion Vector Selection:** Comparing the calculated MAD values:

- $\text{MAD}(0, 0) = 0.75$
- $\text{MAD}(-1, -1) = 3.00$

Since  $\text{MAD}(0, 0) < \text{MAD}(-1, -1)$ , the motion vector  $\mathbf{v} = (0, 0)$  is selected as the preferred motion vector.

## Quick Revision — Module VII

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### Key Formulas at a Glance

|                                  |                                                                                                     |
|----------------------------------|-----------------------------------------------------------------------------------------------------|
| <b>Mean Absolute Difference</b>  | $MAD(i, j) = \frac{1}{N^2} \sum_{x=0}^{N-1} \sum_{y=0}^{N-1}  F_t(x, y) - F_{t-1}(x + i, y + j) $   |
| <b>Mean Squared Error</b>        | $MSE(i, j) = \frac{1}{N^2} \sum_{x=0}^{N-1} \sum_{y=0}^{N-1} [F_t(x, y) - F_{t-1}(x + i, y + j)]^2$ |
| <b>Search Complexity (Full)</b>  | $N_{\text{eval}} = (2p + 1)^2 \quad (p = \text{search range})$                                      |
| <b>Search Complexity (TSS)</b>   | $N_{\text{eval}} \approx 9 + 8 \log_2(p)$                                                           |
| <b>Motion Compensation Error</b> | $e_t(x, y) = F_t(x, y) - F_{t-1}(x + dx, y + dy)$                                                   |

### One-Line Definitions

- **Motion Estimation:** The process of locating the displacement vector of a macroblock between successive video frames.
- **Motion Compensation:** The process of reconstructing a frame by shifting reference frame pixels using motion vectors.
- **I-Frame:** An intra-coded frame compressed independently using spatial details, acting as a random access sync point.
- **P-Frame:** A predicted frame coded using forward temporal prediction from a single preceding reference frame.
- **B-Frame:** A bidirectional frame coded using both past and future reference frames, offering maximum compression.
- **GOP:** Group of Pictures, defining the recurring sequence pattern of I, P, and B frames in a video stream.

### Frequently Asked / Viva Questions

1. **What is temporal redundancy in video coding?**  
It is the high correlation between consecutive frames in a video sequence, where background and moving objects remain mostly identical from frame to frame.
2. **How does the Hybrid Video Encoder combine spatial and temporal compression?**  
It uses motion estimation (temporal prediction) to generate a residual prediction error, which is then compressed using 2D DCT and quantization (spatial transform coding).
3. **Why do B-frames achieve much higher compression than P-frames?**  
Because B-frames use bidirectional prediction, meaning they can reference macroblocks from both past and future frames, yielding smaller prediction errors.
4. **What is the trade-off of using a large number of B-frames in video streaming?**

**Frequently Asked / Viva Questions (contd.)**

B-frames require future frames to be decoded first, which introduces frame buffering delay (latency) and increases decoder memory requirements.

**5. Explain the Three-Step Search (TSS) algorithm.**

It is a fast search method that evaluates 9 points using a coarse step size, shifts the center to the minimum error point, halves the step size, and repeats until the step size is 1.

**6. What is a macroblock in video standards?**

A macroblock is a  $16 \times 16$  pixel region of luminance (Y) along with the corresponding chrominance (U, V) blocks, representing the basic processing unit for motion compensation.

**7. State the main advantage of H.264/AVC over MPEG-2.**

H.264/AVC achieves comparable visual quality at approximately half the bitrate of MPEG-2 by using intra-prediction, multi-frame references, and variable block sizes.

**8. What are CTUs in H.265/HEVC?**

Coding Tree Units (CTUs) replace macroblocks in H.265. They can be dynamically partitioned into Coding Units (CUs) from  $64 \times 64$  down to  $8 \times 8$ , optimizing compression for high-resolution (4K/8K) video.

**9. Why are P-frames and B-frames unsuitable as random access entry points?**

Because they contain temporal prediction residuals, meaning they cannot be decoded without first decoding their past and future reference frames.

**10. Define the Motion Vector.**

The displacement vector ( $dx, dy$ ) representing the horizontal and vertical offset of a matching block in the reference frame relative to the current frame.

**Comparison of Video Frame Types**

| Attribute           | I-Frame (Intra)          | P-Frame (Predicted)         | B-Frame (Bidirectional)  |
|---------------------|--------------------------|-----------------------------|--------------------------|
| Coding Source       | Self-contained (spatial) | Temporal (forward)          | Temporal (bidirectional) |
| Compression Ratio   | Low (2 : 1 to 5 : 1)     | Moderate (10 : 1 to 20 : 1) | High (30 : 1 to 50 : 1)  |
| Random Access Point | Yes                      | No                          | No                       |